BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: MRD-9-[illegible in source scan]090525 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** MRD-9-[illegible in source scan]090525 (A), VERSION: C

Note: There are three sections in this paper i.e A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e A, B, C or D.

Q1. Work done by an elevator is 1200J with a constant force of 400N, distance covered by elevator is:

(A) 2m
(B) 3m
(C) 4m
(D) 6m

Q2. A book placed on the top shelf of cupboard possesses:

(A) Potential energy
(B) Chemical energy
(C) Kinetic energy
(D) Thermal energy

Q3. Atmospheric pressure increases near the surface of earth because:

(A) Density of air Increases
(B) Density of air decreases
(C) Temperature decreases
(D) Strength of gravity decreases

Q4. Which one has high specific heat capacity?

(A) Mercury
(B) Glycerin
(C) Ethanol
(D) Water

Q5. To increase the speed of free electrons in metals we have to increase its:

(A) Length
(B) Temperature
(C) No of electrons
(D) Area of cross section

Q6. Which of the following is derived physical quantity?

(A) Acceleration
(B) Amount of Substance
(C) Electric Current
(D) Temperature

Q7. A car is moving along a curved path, its meter shows 800km/hr which shows its:

(A) Distance
(B) Acceleration
(C) Speed
(D) Velocity

Q8. The maximum static friction on a body just before it starts moving is called:

(A) Limiting friction
(B) Sliding friction
(C) Kinetic friction
(D) Rolling friction

Q9. The centripetal acceleration of an object moving with speed 8m/s in a circle of radius 4m is:

(A) $4ms/s^2$
(B) $8ms/s^2$
(C) $12ms/s^2$
(D) $16ms/s^2$

Q10. The x-component of a force is 20N and Y-component is 15N. What will be the magnitude of force?

(A) 15N
(B) 20N
(C) 25N
(D) 30N

Q11. Bottle opener is the practical application of:

(A) Centripetal force
(B) Centrifugal force
(C) Torque
(D) Momentum

Q12. The gravitational force 'Fg' between two objects does not depend on:

(A) Sum of masses
(B) Product of masses
(C) Gravitational constant
(D) Distance between masses

Section – B

Time Allowed: 2.40 Hours | **Maximum Marks:** 32

Q2. Write short answers to any eight (8) of the following parts. All parts carry equal marks:-

(i) Define any four branches of Physics?

(ii) Define vector quantities. Graphically represent 10N force acting towards South.

(iii) A 6kg object is moving with velocity of 3m/s when it collides with another object resulting in its new velocity of 2m/s. Calculate the change in momentum.

(iv) How centripetal force is used in centrifuge?

(v) Differentiate between like and unlike parallel forces by giving examples?

(vi) What is first condition of equilibrium? Give its mathematical form.

(vii) By giving examples describe renewable and non-renewable energy sources.

(viii) Calculate the power of a 60kg person who climbs 5 meter high tree in 10 seconds.

(ix) Briefly explain stress and strain curve.

(x) Define evaporation. How dryness of air effects the rate of evaporation?

(xi) Why is the cutting edge of the knife made very thin?

Section – C

Maximum Marks: 21

> Note: Answer any three (3) questions. All questions carry equal marks:-

Q3.

(a) Derive 3rd equation of motion using velocity time graph.

(b) A stone is dropped from a bridge reaches the water in 5.30s. Find the height of bridge from water and velocity of stone when it strikes the water.

Q4.

(a) By using the law of gravitation calculate the mass of earth.

(b) Calculate the mass of Jupiter when radius is rj = 7.25 x 10^7m and acceleration due to gravity on this is 24.7 m/s^2.

Q5.

(a) Explain Pascal's Principle.

(b) What is the pressure at a depth of 100m below the surface of water?

Q6.

(a) Define latent heat. Explain specific latent heat of vaporization, give its mathematical form and unit.

(b) How much heat is required to increase the temperature of 2.5kg of water (with specific heat capacity of 4190J/$Kg^{-1}K^{-1}$) from 20°C to 75°C?